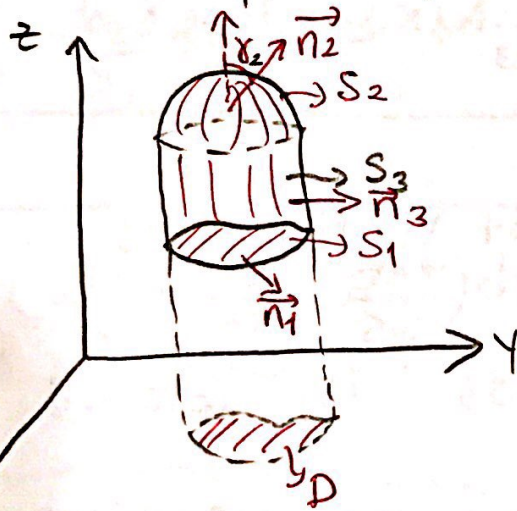


Divergence Theorem (Gauss - Ostrogradski) (16)

Let the surface S be generated by the surface parts S_1, S_2, S_3 , and the projection of this space region V on xy -plane be the region D . Consider the functions $L(x, y, z)$, $M(x, y, z)$ and $N(x, y, z)$ that are defined and continuous on this space region V , also assume that partial derivatives exist.

$$S_1: z_1 = f_1(x, y), \quad S_2: z_2 = f_2(x, y)$$



x Let consider the function $N(x, y, z)$. It's defined and continuous on V .

$$I = \iiint_V \frac{\partial N}{\partial z} dx dy dz$$

$$I = \iint_D \left[\int_{f_1}^{f_2} \frac{\partial N}{\partial z} dz \right] dx dy$$

$$I = \iint_D N(x, y, f_2) dx dy - \iint_D N(x, y, f_1) dx dy$$

for the surface S_2 $\cos(\vec{n}_2, z) > 0$

for the surface S_1 $\cos(\vec{n}_1, z) < 0$

for the surface S_3 $\cos(\vec{n}_3, z) = 0$
 $(\vec{n}_3 \perp z, \gamma = \pi/2)$

$$\iint_D N(x, y, f_2) \underbrace{dx dy}_{\cos \gamma ds = dx dy} = \iint_{S_2} N(x, y, z) \cos \gamma_2 ds$$

$$+ \iint_D N(x, y, f_1) dx dy = - \iint_{S_1} N(x, y, z) \cos \gamma_1 ds$$

$$0 = \iint_{S_3} N(x, y, z) \cos \gamma_3 ds \text{ (add)}$$

+

$$I = \iint_{S_1} N \cos \gamma_1 ds + \iint_{S_2} N \cos \gamma_2 ds + \iint_{S_3} N \cos \gamma_3 ds$$

$$\iiint_V \frac{\partial N}{\partial z} dz dx dy = \iint_S N \cos \gamma ds \text{ is obtained.}$$

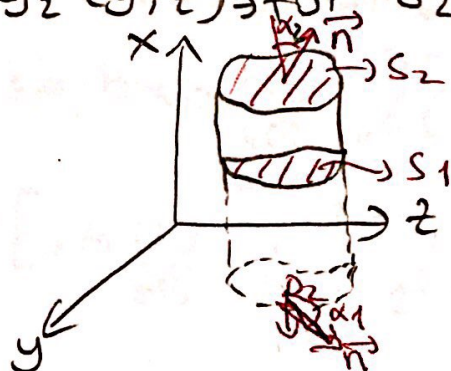
⊗ Similarly; have projection on yz -plane; (D_2)

$$I = \iiint_V \frac{\partial L}{\partial x} dv, \quad \left\{ \begin{array}{l} x = g_1(y, z) \rightarrow \text{for } S_1 \\ x = g_2(y, z) \rightarrow \text{for } S_2 \end{array} \right\}$$

$$I = \iint_{D_2} \int_{x_1}^{x_2} \frac{\partial L}{\partial x} dx dy dz$$

$$I = \iint_{D_2} L(x, y, z) \Big|_{g_1(y, z)}^{g_2(y, z)} dy dz$$

$$I = \iint_{D_2} L(g_2(y, z), y, z) dy dz - \iint_{D_2} L(g_1(y, z), y, z) dy dz$$



$$ds = \sec \alpha dy dz$$

$$dy dz = \cos \alpha ds$$

$$I = \iint_{S_2} L(x, y, z) \cos \alpha_2 ds + \iint_{S_1} L(x, y, z) \cos \alpha_1 ds$$

$$I = \iint_S L(x, y, z) \cos \alpha ds$$

$$\iiint_V \frac{\partial N}{\partial z} dv = \iint_S N(x, y, z) \cos \gamma ds$$

$$\iiint_V \frac{\partial L}{\partial x} dv = \iint_S L(x, y, z) \cos \alpha ds$$

$$\iiint_V \frac{\partial M}{\partial y} dv = \iint_S M(x, y, z) \cos \beta ds \quad \left(\begin{array}{l} \text{can be obtained} \\ \text{by the projection} \\ \text{on } xz\text{-plane} \end{array} \right)$$

+

$$\iiint_V \underbrace{\left(\frac{\partial N}{\partial z} + \frac{\partial L}{\partial x} + \frac{\partial M}{\partial y} \right)}_{\text{div } \vec{F}} dv = \iint_S (N \cos \gamma + L \cos \alpha + M \cos \beta) ds$$

$$\vec{F} = L \vec{i} + M \vec{j} + N \vec{k}$$

$$\vec{N} = \cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k}$$

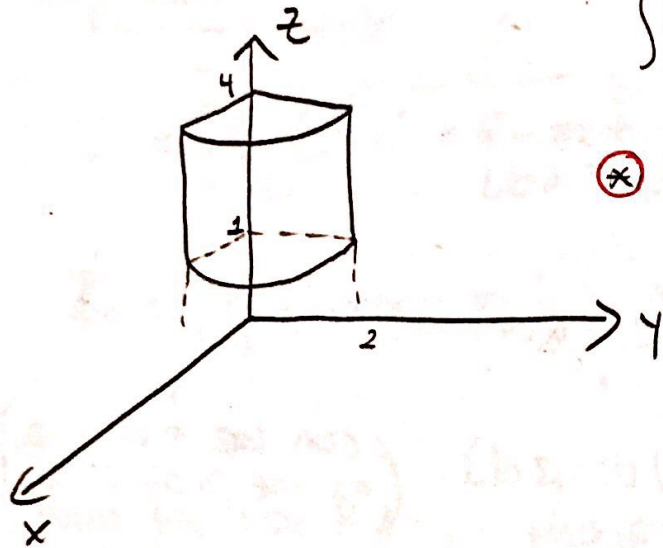
$$\boxed{\iiint_V \text{div } \vec{F} dv = \iint_S \vec{F} \cdot \vec{N} ds}$$

Divergence Formula

Example: $S: \begin{cases} x^2 + y^2 = 4 \\ x=0, y=0, z=1, z=4 \end{cases}$ and

$\vec{F} = y^2 \vec{i} - x \vec{j} + xz \vec{k}$ are given. Verify Divergence Formula.

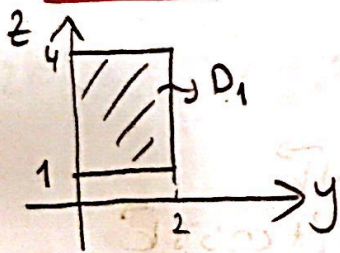
$$\iiint_V \text{div} \vec{F} dv = \iint_S \vec{F} \cdot \vec{N} ds$$



⊗ There are 5 surfaces:

$$\begin{aligned} S_1 &\rightarrow x=0 \\ S_2 &\rightarrow y=0 \\ S_3 &\rightarrow z=1 \\ S_4 &\rightarrow z=4 \\ S_5 &\rightarrow x^2 + y^2 = 4 \end{aligned}$$

For S_1 : $x=0$, projection on yz -plane



$$\begin{aligned} \vec{N} &= -\vec{i} \\ ds &= \sec \alpha dy dz \\ ds &= dy dz \quad (\because \sec \alpha = 1) \end{aligned}$$

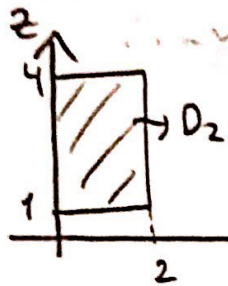
$$\left(\sec \alpha = \sqrt{\left(\frac{\partial x}{\partial y} \right)^2 + \left(\frac{\partial x}{\partial z} \right)^2 + 1} \right)$$

$$I_1 = \iint_{S_1} \vec{F} \cdot \vec{N} ds = \iint_{S_1} (y^2 \vec{i} - x \vec{j} + xz \vec{k}) \cdot (-\vec{i}) dy dz$$

$$I_1 = \iint_{D_1} -y^2 dy dz = \int_1^4 \int_0^2 -y^2 dy dz = \int_1^4 \left[-\frac{y^3}{3} \right]_0^2 dz = -\frac{1}{3} \int_1^4 8 dz$$

$$I_1 = -\frac{1}{3} \cdot 8 \cdot (4-1) = -8$$

For S_2 : $y=0$, projection on xz -plane



$$\vec{N} = -\vec{j}, \quad ds = \sec \beta dx dz$$

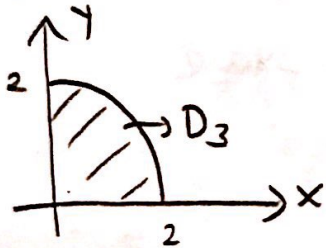
$$ds = dx dz$$

$$(* \sec \beta = 1)$$

$$I_2 = \iint_{S_2} (\underbrace{y^2 \vec{i} - x \vec{j} + xz \vec{k}}_{\vec{F}}) (\underbrace{-\vec{j}}_{\vec{N}}) \underbrace{dx dz}_{ds} = \iint_{S_2} x dx dz$$

$$I_2 = \int_1^4 \int_0^2 x dx dz = \frac{1}{2} \int_1^4 x^2 \Big|_0^2 dz = \frac{1}{2} \int_1^4 4 dz = \frac{1}{2} \cdot 4 \cdot 3 = 6 //$$

For S_3 : $z=1$, projection on xy -plane;



$$\vec{N} = -\vec{k}, \quad ds = \sec \gamma dx dy \quad (* \sec \gamma = 1)$$

$$ds = dx dy$$

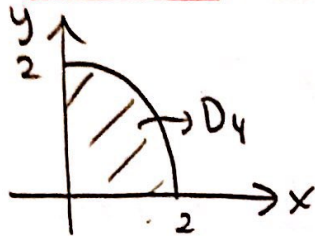
$$I_3 = \iint_{S_3} (\underbrace{y^2 \vec{i} - x \vec{j} + xz \vec{k}}_{\vec{F}}) (\underbrace{-\vec{k}}_{\vec{N}}) \underbrace{dx dy}_{ds} = \iint_{D_3} \underbrace{-xz}_{\downarrow 1 \quad (*z=1)} dx dy$$

$$I_3 = - \iint_{D_3} x dx dy = - \int_0^{\pi/2} \int_0^2 r \cos \theta \cdot r dr d\theta, \quad * \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$I_3 = - \int_0^{\pi/2} \int_0^2 r^2 \cos \theta dr d\theta = - \frac{1}{3} \int_0^{\pi/2} r^3 \Big|_0^2 \cos \theta d\theta$$

$$I_3 = - \frac{8}{3} \int_0^{\pi/2} \cos \theta d\theta = - \frac{8}{3} \cdot \sin \theta \Big|_0^{\pi/2} = - \frac{8}{3} (1 - 0) = - \frac{8}{3} //$$

For S_4 : $z=4$, projection on xy -plane



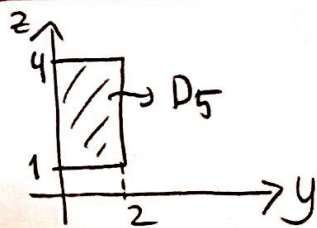
$$\vec{N} = \vec{k}, \quad ds = \sec \gamma dx dy \quad (\sec \gamma = 1) \\ ds = dx dy$$

$$I_4 = \iint_{S_4} (\overbrace{y^2 \vec{i} - x \vec{j} + xz \vec{k}}^{\vec{F}}) (\overbrace{\vec{k}}^{\vec{N}}) \overbrace{dx dy}^{ds} = \iint_{D_4} x \cdot \underbrace{z}_{=4} dx dy$$

$$I_4 = 4 \iint_{D_4} x dx dy = 4 \int_0^{\pi/2} \int_0^2 r \cos \theta \cdot r dr d\theta = 4 \int_0^{\pi/2} \int_0^2 r^2 \cos \theta dr d\theta$$

$$I_4 = \frac{4}{3} \int_0^{\pi/2} r^3 \Big|_0^2 \cos \theta d\theta = \frac{32}{3} \int_0^{\pi/2} \cos \theta d\theta = \frac{32}{3} \sin \theta \Big|_0^{\pi/2} = \frac{32}{3} //$$

For S_5 : $x^2 + y^2 = 4$, projection on yz -plane;



$$\vec{N} = \frac{2x \vec{i} + 2y \vec{j}}{\sqrt{4x^2 + 4y^2}} = \frac{x \vec{i} + y \vec{j}}{\sqrt{x^2 + y^2}} = \frac{x \vec{i} + y \vec{j}}{2}$$

$$\vec{N} \cdot \vec{i} = \frac{x}{2} = \cos \alpha \Rightarrow \sec \alpha = \frac{2}{x}, \quad ds = \frac{2}{x} dy dz$$

$$I_5 = \iint_{S_5} (\overbrace{y^2 \vec{i} - x \vec{j} + xz \vec{k}}^{\vec{F}}) \cdot \underbrace{\left(\frac{x \vec{i} + y \vec{j}}{2} \right)}_{\vec{N}} \cdot \underbrace{\frac{2}{x}}_{ds} dy dz$$

$$I_5 = \iint_{D_5} \left(\frac{y^2 x - xy}{x} \right) dy dz = \iint_{D_5} (y^2 - y) dy dz$$

$$I_5 = \int_1^4 \int_0^2 (y^2 - y) dy dz = \int_1^4 \left(\frac{y^3}{3} - \frac{y^2}{2} \right) \Big|_0^2 dz = \int_1^4 \left(\frac{8}{3} - 2 \right) dz$$

$$I_5 = \frac{2}{3} \cdot (4-1) = 2 //$$

$$I = \iint_S \vec{F} \cdot \vec{n} ds = -8 + 6 - \frac{8}{3} + \frac{32}{3} + 2 = 8 //$$

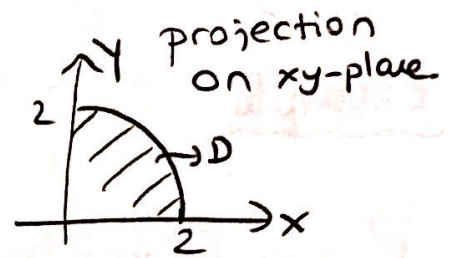
* We need to find $\iiint_V \text{div } \vec{F} \, dv = ?$

$$\vec{F} = y^2 \vec{i} - x \vec{j} + xz \vec{k}$$

$$\text{div } \vec{F} = \frac{\partial}{\partial x}(y^2) + \frac{\partial}{\partial y}(-x) + \frac{\partial}{\partial z}(xz)$$

$$\text{div } \vec{F} = 0 + 0 + x \Rightarrow \boxed{\text{div } \vec{F} = x}$$

$$I = \iiint_V x \, dv = \iiint_V x \, dz \, dx \, dy$$



$$I = \iint_D \int_{z=1}^{z=4} x \, dz \, dx \, dy = \iint_D x \cdot z \Big|_1^4 \, dx \, dy = 3 \iint_D x \, dx \, dy$$

$$I = 3 \int_0^{\pi/2} \int_0^2 r \cos \theta \cdot r \, dr \, d\theta = \int_0^{\pi/2} r^3 \Big|_0^2 \cos \theta \, d\theta$$

$$I = 8 \int_0^{\pi/2} \cos \theta \, d\theta = 8 \cdot \sin \theta \Big|_0^{\pi/2} = 8(1-0) = 8 //$$

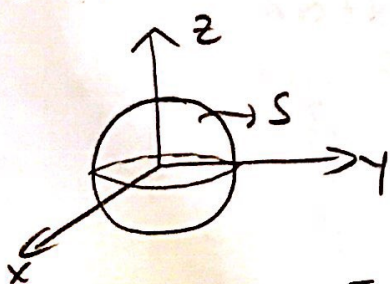
Example: Evaluate the integral I

$$I = \iiint_S x \, dy \, dz + y \, dx \, dz + z \, dx \, dy \text{ on } S: x^2 + y^2 + z^2 = 1.$$

$$\vec{F} = x \vec{i} + y \vec{j} + z \vec{k}, \quad \vec{n} = \cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k}$$

According to divergence Theorem; $\iint_S \vec{F} \cdot \vec{n} \, ds = \iiint_V \text{div } \vec{F} \, dv$

$$\text{div } \vec{F} = 1 + 1 + 1 = 3$$



$$I = \iiint_V 3 \, dz \, dx \, dy = \int_0^{2\pi} \int_0^{\pi} \int_0^1 3r^2 \sin \phi \, dr \, d\phi \, d\theta$$

$$I = 3 \int_0^{2\pi} \int_0^{\pi} \sin \phi \cdot r^3 \Big|_0^1 \, d\phi \, d\theta$$

$$I = \int_0^{2\pi} \int_0^{\pi} \sin \phi \, d\phi \, d\theta = \int_0^{2\pi} -\cos \phi \Big|_0^{\pi} \, d\theta$$

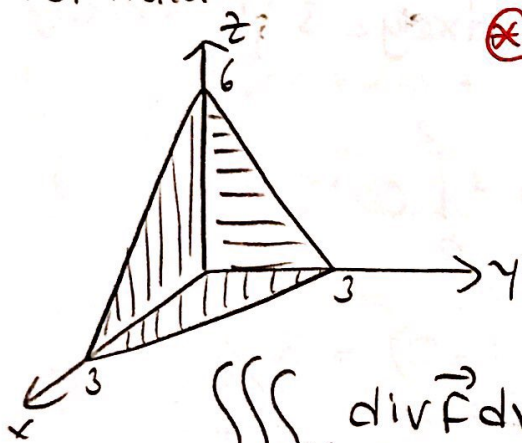
$$I = \int_0^{2\pi} -\cos \theta \Big|_0^{2\pi} d\theta = - \int_0^{2\pi} (\underbrace{\cos 2\pi}_{-1} - \underbrace{\cos 0}_1) d\theta$$

$$I = - \int_0^{2\pi} (-2) d\theta = 4\pi //$$

Example: $S: \begin{cases} 2x+2y+z=6 \\ x=0, y=0, z=0 \end{cases}$ and

$\vec{F} = xy\vec{i} - x^2\vec{j} + (x+z)\vec{k}$ are given. Verify Divergence Formula.

⊗ There are 4 surfaces;

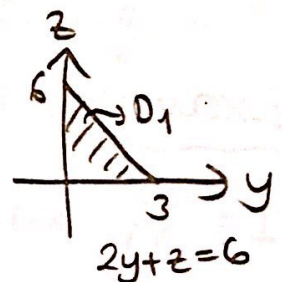


$$\left. \begin{aligned} S_1 &\rightarrow x=0 \\ S_2 &\rightarrow y=0 \\ S_3 &\rightarrow z=0 \\ S_4 &\rightarrow 2x+2y+z=6 \end{aligned} \right\}$$

$$\iiint_V \text{div} \vec{F} dv = \iint_S \vec{F} \cdot \vec{N} ds$$

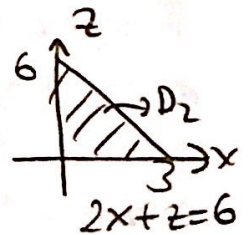
For S_1 : $x=0$, $\vec{N} = -\vec{i}$, $ds = \sec \alpha dy dz$
 $ds = dy dz$

$$I_1 = \iint_{S_1} \vec{F} \cdot \vec{N} ds = \iint_{D_1} -xy dy dz = 0 //$$



For S_2 : $y=0$, $\vec{N} = -\vec{j}$, $ds = \sec \beta dx dz$

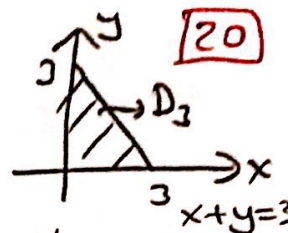
$$I_2 = \iint_{S_2} \vec{F} \cdot \vec{N} ds = \iint_{D_2} x^2 dx dz = \int_0^3 \int_0^{6-2x} x^2 dz dx$$



$$I_2 = \int_0^3 x^2 \cdot z \Big|_0^{6-2x} dx = \int_0^3 x^2 (6-2x) dx = \int_0^3 (6x^2 - 2x^3) dx ;$$

$$I_2 = 2x^3 - \frac{1}{2}x^4 \Big|_0^3 = 2 \cdot 27 - \frac{81}{2} = \frac{27}{2} //$$

For S_3 : $z=0$, $\vec{N} = -\vec{k}$, $ds = \sec \gamma dx dy$
 $ds = dx dy$



$$I_3 = \iint_{S_3} \vec{F} \cdot \vec{N} ds = \iint_{S_3} -(x+z) dx dy = - \iint_{D_3} x dx dy$$

$$I_3 = - \int_0^3 \int_0^{3-y} x dx dy = - \frac{1}{2} \int_0^3 x^2 \Big|_0^{3-y} dy = - \frac{1}{2} \int_0^3 (9 - 6y + y^2) dy$$

$$I_3 = - \frac{1}{2} \left(9y - 3y^2 + \frac{y^3}{3} \right) \Big|_0^3 = - \frac{1}{2} \left(\cancel{27} - \cancel{27} + \frac{27}{3} \right) = - \frac{9}{2} //$$

For S_4 : $2x+2y+z=6$ projection on xy

$$\vec{N} = \frac{2\vec{i} + 2\vec{j} + \vec{k}}{3}, \quad \vec{n} \cdot \vec{k} = \frac{1}{3} = \cos \gamma$$

$$\sec \gamma = 3 \Rightarrow ds = 3 dx dy$$

$$I_4 = \iint_{S_4} \vec{F} \cdot \vec{N} ds = \iint_{D_4} [xy\vec{i} - x^2\vec{j} + (x+z)\vec{k}] \cdot \left[\frac{2\vec{i} + 2\vec{j} + \vec{k}}{3} \right] 3 dx dy$$

$$I_4 = \iint_{D_4} (2xy - 2x^2 + x + z) dx dy$$

$$I_4 = \iint_{D_4} (2xy - 2x^2 + x + 6 - 2x - 2y) dx dy$$

$$I_4 = \int_0^3 \int_0^{3-y} (2xy - 2x^2 - x - 2y + 6) dx dy$$

$$I_4 = \int_0^3 \left(x^2 y - \frac{2}{3} x^3 - \frac{x^2}{2} - 2xy + 6x \right) \Big|_0^{3-y} dy$$

$$I_4 = \int_0^3 \left((3-y)^2 y - \frac{2}{3} (3-y)^3 - \frac{(3-y)^2}{2} - 2(3-y)y + 6(3-y) \right) dy$$

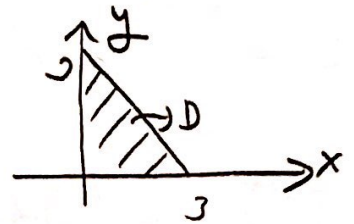
$$I_4 = 27/4$$

$$I = \iint_S \vec{F} \cdot \vec{N} ds = 0 + \frac{27}{2} - \frac{9}{2} + \frac{27}{4} = \frac{63}{4} //$$

⊗ We need to find $\iiint_V \text{div} \vec{F} dv = \frac{63}{4} ??$

$$\vec{F} = xy\vec{i} - x^2\vec{j} + (x+z)\vec{k}$$

$$\text{div} \vec{F} = y+1, \quad I = \iiint_V (y+1) dv$$



$$I = \iint_D \int_{z=0}^{z=6-2x-2y} (y+1) dz dx dy = \iint_D (6-2x-2y)(y+1) dx dy$$

$$I = \iint_D (6y - 2xy - 2y^2 + 6 - 2x - 2y) dx dy$$

$$I = \iint_D (4y - 2xy - 2y^2 - 2x + 6) dx dy$$

$$I = \int_0^3 (4xy - x^2y - 2y^2x - x^2 + 6x) \Big|_0^{3-y} dy$$

$$I = \int_0^3 [4y(3-y) - y(3-y)^2 - 2y^2(3-y) - (3-y)^2 + 6(3-y)] dy$$

$$I = \int_0^3 (\underline{12y} - \underline{4y^2} - y(9 - 6y + y^2) - \underline{6y^2} + \underline{2y^3} - (9 - 6y + y^2) + \underline{18} - \underline{6y}) dy$$

$$I = \int_0^3 (\underline{6y} - \underline{10y^2} + \underline{2y^3} + \underline{18} - \underline{9y} + \underline{6y^2} - \underline{y^3} - \underline{9} + \underline{6y} - \underline{y^2}) dy$$

$$I = \int_0^3 (y^3 - 5y^2 + 3y + 9) dy = \left[\frac{y^4}{4} - \frac{5}{3}y^3 + \frac{3}{2}y^2 + 9y \right]_0^3$$

$$I = \frac{81}{4} - \frac{5}{3} \cdot 27 + \frac{3}{2} \cdot 9 + 27 = \underline{\underline{63/4}}$$